Instructor: Prof. Amini
Date: 1404/___/___

Sharif University
of Technology

Advanced Traffic (20–551)
Class Project Part 4

Part 4: Simulation Engine & Data Logging

Goals:

1. Use `SimPy` or a custom loop to implement the simulation engine.
2. Read the demand file and generate vehicles correctly.
3. Build the simulation loop and collect data.

Tasks

1. Implement a simulation engine (using `SimPy` or a custom loop).
2. Generate vehicles from demand data and move them according to the defined logic.
3. Log vehicle positions and states to a `DataFrame`.

Output

1. Implement `Engine.py` to run the simulation.
2. Modify Python modules as needed to integrate with `Engine.py`.
3. Provide brief documentation in all files and include a `README.md`.
4. Present the code and answer questions.
5. Provide the log file (CSV).

Time

Weeks 3–4.

Grade Weight

40% total for Parts 2–4 [Part 2: 20%, Part 3: 10%, Part 4: 10%].
- If there are problems in the vehicles or network, you will lose a portion of the 20%.
- If there are problems in the traffic light logic, you will lose a portion of the 10%.
- If there are problems in the Engine, you will likely lose the 10% for Part 4.